

Wayner Barrios, Ph.D.

AI Research Scientist — Multimodal LLMs & Computer Vision

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Summary

Ph.D. in Computer Science (Dartmouth, completed Winter 2026), specializing in multimodal AI. Research on multimodal reasoning, video understanding, and cross-modal fusion, published at top-tier venues. Shipped production ML systems for healthcare (4+ hospitals), defense (IARPA/DARPA), and enterprise clients across US & LATAM. Open-source contributor; built inference tools for Apple Silicon and Blackwell architectures. Skilled in PyTorch/JAX, distributed training, and large-scale data pipelines.

Core Expertise

Research Multimodal LLMs • Long-video understanding • MLLM evaluation & benchmarking • Cross-modal attention/fusion • Multimodal reasoning

Engineering Python, C++ • PyTorch, JAX, MLX • GPU acceleration (CUDA/ROCm) • Distributed training (FSDP, DeepSpeed) • Model optimization (Quantization, LoRA, distillation)

Infrastructure Kubernetes, Docker • AWS & GCP • Vector stores & retrieval pipelines • High-throughput data systems

Education

Ph.D. in Computer Science 2021–2026

Dartmouth College – Hanover, NH

Visual Computing Lab & Machine Learning Lab. Advisor: Prof. SouYoung Jin

Research: Multimodal LLMs, step-verified reasoning, video understanding

Visiting Researcher 2018–2019

Carnegie Mellon University – Robotics Institute, Pittsburgh, PA

Human Sensing Lab & DIVA Project. Funded by IARPA/NIST. Supervisors: F. de la Torre, D. Huang, A. Hauptmann

Visiting Student Researcher 2016

KAUST – Saudi Arabia. Image and Video Understanding Lab. Supervisor: Prof. Bernard Ghanem

B.S. in Electronics Engineering 2010–2015

Universidad del Norte – Colombia. Ranked 1st in EE Department, 2nd overall

Research & Industry Experience

Founder 2018–Present

Wiqonn – Remote

LATAM-based AI consultancy; data analytics, ML pipelines, and research-backed solutions for enterprise clients.

AI Consultant 2024–2025

GeoSure AI – Remote

Built agentic LLM pipelines for real-time location-based risk assessment from news/social signals. Delivered production dashboards for risk scoring.

AI Research Intern Jan–Mar 2025

Samsung Research America – Mountain View, CA

Video reasoning models with GRPO-based RL for alignment. Improved benchmark performance by 15%; integrated into internal product.

Graduate Researcher — Multimodal AI 2021–2026

Dartmouth College – Hanover, NH

Designed and evaluated multimodal reasoning systems under Prof. SouYoung Jin. Developed CRYSTAL, a benchmark for evaluating stepwise reasoning in vision-language models. Built scalable pipelines for atomic step generation with semantic clustering and multi-stage validation. Implemented RL training (PPO/GRPO) with composite rewards for step-matching. Improved long-video moment retrieval via multimodal guidance and hard-negative mining.

Co-Researcher & Technical Lead 2020–2021

VivaMed - EAFIT University – Colombia

Deployed COVID-19 detection in 4+ hospitals; federated learning for HIPAA compliance; 90%+ accuracy. Funded by Colombian Ministry of Science.

Software Engineer & Data Scientist 2016–2018

Piensa Labs – Remote — Clients: Harvard CGA, World Bank, Voyage (Cruise), Max Planck

Harvard: Geospatial architecture for WorldMap; tile generation, spatial indexing; ML for Billion Object Platform. World

Bank: Data integration for risk analytics; GeoNode open-source contributor. Voyage: CV annotation tools for autonomous driving. Max Planck: Curated PoseTrack benchmark.

Research Assistant

2015–2016

ActivityNet Project – KAUST & Universidad del Norte

Co-created ActivityNet v1.2/v1.3 benchmarks; designed QA pipeline for temporal annotations → CVPR 2017

Open Source

vLLM-MLX: OpenAI-compatible inference server for Apple Silicon; continuous batching, MCP tools, multimodal support.

DGX Spark Fine-tune: LoRA + NVFP4/MXFP8 quantization for NVIDIA Blackwell GB10.

GeoNode: Core contributor; service virtualization for open-source geospatial platform.

ActivityNet: Co-creator of v1.2/v1.3, industry-standard video understanding benchmarks.

Selected Publications

[arXiv'26] **W. Barrios**. “Native LLM and MLLM Inference at Scale on Apple Silicon.”

[arXiv'25] **W. Barrios**, et al. “MoDA: Modulation Adapter for Fine-Grained Visual Grounding in Instructional MLLMs.”

[WACV'25] X. Diao, M. Cheng, **W. Barrios**, S. Jin. “FT2TF: First-Person Statement Text-To-Talking Face Generation.”

[arXiv'24] **W. Barrios**, S. Jin. “Multi-layer Learnable Attention Mask for Multimodal Tasks.”

[ICCV'23] **W. Barrios**, et al. “Localizing Moments in Long Video Via Multimodal Guidance.”

[FMT'22] A. Murillo-González, et al. (incl. **W. Barrios**). “Medical decision support using weakly-labeled lung CT scans.”

[JPI'22] **W. Barrios**, et al. “Bladder cancer prognosis using deep neural networks and histopathology images.”

[CVPR'17] F. Caba Heilbron, **W. Barrios**, et al. “SCC: Semantic Context Cascade for Efficient Action Detection.”

Full list: Google Scholar

Leadership & Service

Program Chair: ActivityNet Challenge @ CVPR 2016. **Mentorship Chair:** LatinX in AI @ ICML/ECCV 2022.

Reviewer: CVPR, ECCV, NeurIPS, ICLR, ICCV, ACL, AAAI.

Teaching: 7+ graduate courses (ML, Generative AI, Data Science, Databases).

Competition: 2nd place, Activities in Extended Videos Challenge @ CVPR 2019 (CMU team).

Awards

Dartmouth Graduate Fellowship (2021–2026)

CMU Robotics Institute Fellowship (2018)

KAUST Visiting Scholar Award (2016)

COVID-19 Research Grant (2020)

Silver Medal Academic Excellence (2015)